

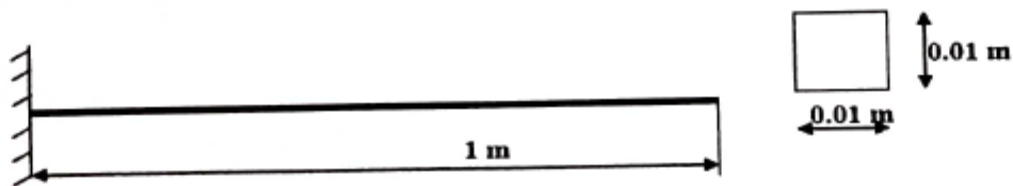
Practical 8

Title – Model Analysis of Cantilever Beam

Problem Statement

Perform the Modal Analysis of Cantilever beam for natural frequency determination. Modulus of Elasticity $E=2 \times 10^5 \text{ N/mm}^2$, Density = 7800 Kg/m^3 ,

Poisson's ratio = 0.27



Procedure

Step 1 Preferences

Open Ansys APDL – Preferences – Structural – OK

Step 2 Pre-processor

Pre Processor → Add/Edit/Delete → Add → Beam → 2 node 188 → Ok

Step 3 Material

Material properties → Material Models → Structural → Linear → Elastic →

Isotropic → Add $E=2e5$ → $\nu=0.27$ → OK → Density → 7800 → OK

Step 4 Sections

Sections – Beam - Common Sections - Enter B 0.01 & H 0.01 - OK

Step 5 Modeling

Modeling – Create –Key point 1 – In Active CS - 0,0,0 – Apply – Key point

2 – 1, 0, 0 – OK

Modeling – Create – Lines – Straight Lines – Pick Key point 1 and 2 - OK

Step 6 Meshing

Meshing – Size Controls – Manual Size – Lines – All Lines – No of Element

Division - 100 – Ok

Mesh – Lines – Pick All

Step 7 Loads

Loads – Define Loads –Apply – Structural – Displacement – Key point 1 –

Ok – All DOF - Ok

Step 8 Solution

Solution – Analysis Type – New Analysis – Modal - OK

Solution – Analysis Type – Sub Space – Analysis Options – No of modes to

extract – 5 – No of modes to expand – 5 – OK – (Use default values) - OK

Solution – Solve –Current LS - OK

Step 9 General Results

Read Results → First Set

Plot Results →Deformed Shape →def+ undeformed →ok.

Plot results → Animate →Deformed Shape →def+ undeformed →ok.

Read Results → Next Set

Plot Results →Deformed Shape →def+ undeformed →ok.

Plot results → Animate →Deformed Shape →def+ undeformed →ok.